CLIO Search: Agentic Search for Scientific Data

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***Abstract*—Scientific data search requires human reasoning that does not scale: inspecting corpora, adapting to metadata quality, converting between unit conventions, refining iteratively. Automating it demands agentic search, yet three gaps make this hard: dimensional blindness (200 kPa must match 200000 Pa through arithmetic no system performs), metadata heterogeneity (datasets range from richly annotated to bare files), and storage fragmentation (data spans backends no engine federates). We present CLIO Search, which closes all three through an observe-decide-act-evaluate cycle. Science-aware operators execute deterministic SI conversion and formula normalization alongside BM25 and dense search. CLIO inspects what metadata each corpus provides and adapts accordingly. Federated search spans heterogeneous HPC backends with agentic refinement. CLIO achieves 0.99 P@5 on cross-unit queries versus 0.40 for the strongest baseline, improves precision 87% on real NOAA data, and returns results in under 50 ms across 2.5M documents at distributed scale. All code and benchmarks will be released.**

I. Introduction

When a scientist searches for data across a large repository, they do not simply type a query and read results. They inspect what each dataset contains, determine whether it has structured metadata or only raw files, decide which search approach will work for each dataset, apply domain knowledge to interpret measurements across different conventions, and refine their search when initial results are insufficient. This process is reasoning, not retrieval. It is how a human expert navigates heterogeneous scientific data. No retrieval system replicates it.

At HPC scale, human reasoning does not scale. A single exascale job on Frontier can produce 80 PB of simulation output [1]. A commonly cited estimate holds that 60 to 80% of scientific data is never reused, not for lack of access, but for lack of search [2]. The FAIR data principles [3], now mandated by NSF, NIH, and DOE [4], require that data be Findable, Accessible, Interoperable, and Reusable, yet automated assessments show no major repository achieves 50% reusability compliance. The retrieval infrastructure that serves these scientists treats all content as plain text.

AI agents are now entering this space. Coscientist [5] executes lab synthesis autonomously, AI Scientist v2 [6] generates research end-to-end, and autonomous agents at ORNL [7] orchestrate experiments across national facilities. But ScienceAgentBench [8] reports 32.4% end-to-end task success, with failures traced to data handling rather than reasoning. A comprehensive survey [9] identifies scientific data management as explicitly underexplored. These agents inherit the same retrieval infrastructure that fails human scientists: the retrieval layer underneath them is blind to scientific structure.

Current retrieval systems fall short of scientific data discovery in three fundamental ways.

First, scientific search requires domain reasoning that retrieval systems do not perform, a gap we call *dimensional blindness*. A query for pressures between 190 and 360 kPa must recognize that "200000 Pa" and "0.2 MPa" are both matches, but this requires arithmetic, not text similarity. Formulas compound the problem: "$F = ma$," "$F = m \cdot a$," and "$ma = F$" express the same physical law but produce distinct token sequences that defeat both keyword and embedding retrieval. The Numeracy Gap study [10] evaluates 13 embedding models and reports 0.54 binary retrieval accuracy on numerical content. String normalization [11] cannot bridge SI prefixes. Learned approaches [12], [13] achieve up to 65% recall improvement but provide no correctness guarantee. The consequences are real: NASA's $327M Mars Climate Orbiter was lost because one system used pound-seconds while another used newton-seconds [14]; the Gimli Glider incident stemmed from metric vs. imperial fuel calculations [15]; and medication errors, including unit conversion mistakes, cause thousands of deaths annually in US healthcare [16].

Second, scientific datasets exhibit extreme *metadata heterogeneity*, and no search system adapts to this variation. Automated FAIR metadata assessments consistently find that fewer than half of datasets in major repositories carry structured, machine-readable annotations [2]. An HDF5 file with Climate and Forecast convention attributes provides explicit units for every variable; a raw binary dump on a parallel filesystem may offer nothing beyond a filename. A fixed retrieval pipeline cannot handle both: it either wastes computation on datasets with no extractable measurements or misses opportunities when rich metadata is available. A search system that profiles each corpus before querying, determining what metadata exists and which strategies are viable, would address this gap, but no current retrieval system provides this capability.

Third, *storage fragmentation* across heterogeneous backends places scientific data beyond the reach of any single search engine. A single experimental campaign may deposit raw output on a parallel filesystem, preprocessed arrays in HDF5, derived features in a vector database, and catalog entries on an external platform. RAGRoute [17] routes queries to relevant data sources via a learned classifier, but searches only over unstructured text. SMURF [18] federates retrieval across scientific repositories by aligning embeddings, but treats data as opaque vectors without inspecting structured metadata. Neither uses metadata to direct search with domain-aware operators. The difficulty is that each backend exposes different

metadata: an HDF5 file carries CF-convention unit attributes, a filesystem offers only filenames, and a vector database stores embeddings with no unit semantics. A federated search system must adapt its operators to what each backend can provide. No existing system combines domain-aware operators, corpus-adaptive orchestration, and federated HPC access.

These challenges share a common requirement: the search system must reason about *how* to search before it executes the search. It must observe what each corpus contains, decide which strategies to apply, act with domain-specific operators, and evaluate whether to refine. This observe-decide-act-evaluate cycle is what separates agentic search from conventional retrieval. It is how human scientists search. No retrieval system implements it.

We present CLIO Search, a retrieval system that implements this cycle for scientific data. CLIO profiles each corpus to discover what metadata exists and selects search strategies based on what the corpus can support. It executes science-aware operators, dimensional conversion and formula normalization, alongside standard text and vector search, and federates across heterogeneous HPC storage backends. An agentic refinement loop expands, narrows, or pivots queries based on intermediate results. Domain knowledge enters the pipeline as verifiable computation: the fact that 5 km = 5000 m is an algebraic invariant, guaranteed correct by construction. We evaluate CLIO on controlled benchmarks, real observatory data, and distributed HPC infrastructure at scale. CLIO achieves 0.99 P@5 on cross-unit queries versus 0.40 for the strongest text baseline. On real NOAA observatory data, precision improves by 87%. At 2.5M arXiv abstracts across 4 GH200 nodes on NCSA DeltaAI, cross-unit queries return in under 50 ms. Our contributions are as follows:

1) **Science-aware retrieval operators.** Dimensional conversion and preliminary formula normalization ($F{=}ma$ vs. $ma{=}F$) execute as deterministic operators alongside BM25 and dense search. A unit registry converts measurements to SI base units through arithmetic multiplication, guaranteeing correctness by construction. The registry covers 58 units across 12 SI domains[1] and extends with one configuration entry per unit.
2) **Corpus-adaptive search orchestration.** Before querying, CLIO inspects what metadata each corpus provides and selects search strategies accordingly. Corpora with structured measurements activate science-aware operators; corpora with no extractable metadata fall back to text retrieval.
3) **Federated scientific search.** A connector architecture spans heterogeneous backend types (filesystems, object stores, HDF5, NetCDF, IOWarp, external catalogs), each declaring its capabilities so the system adapts its operators to match. An agentic multi-hop loop refines queries across backends based on intermediate results.

[1]Length, mass, time, temperature, pressure, velocity, frequency, radioactivity, energy, power, area, acceleration.

II. Background and Related Work

The observe-decide-act-evaluate cycle that CLIO implements requires foundations from multiple areas. We build on hybrid retrieval (Sec. 2.1), are motivated by the numerical reasoning crisis in retrieval and language models (Sec. 2.2), position against quantity-aware retrieval systems that attempt but do not achieve arithmetic conversion (Sec. 2.3), and survey the scientific data discovery landscape in HPC and agentic systems (Sec. 2.4).

A. Hybrid Retrieval Foundations

Modern retrieval combines lexical and dense signals. BM25 [19] remains the strongest lexical baseline; dense retrieval via learned bi-encoders [20] captures semantic similarity that keyword matching misses. Bruch et al. [21] show that a learned convex combination consistently outperforms reciprocal rank fusion (RRF) on TREC benchmarks, establishing the analytic framework we adopt. At the agentic frontier, A-RAG [22] introduces hierarchical retrieval interfaces that let LLM agents compose retrieval strategies dynamically.

We build *on* hybrid retrieval, not beside it. Our science-aware operators execute as parallel branches alongside BM25 and dense search within the same fusion pipeline, adding domain-specific recall without replacing any existing component.

B. The Numerical Reasoning Crisis

Numbers are fundamentally broken in both retrieval systems and large language models. The Numeracy Gap study [10] evaluates 13 embedding models on numerical content and reports 0.54 binary retrieval accuracy, barely above random chance. NC-Retriever [13] finds that dense retrieval achieves only 16.3% accuracy on queries with numeric constraints, even when the constraint is stated explicitly in both query and document. The root cause is architectural: tokenizers encode numbers per-digit rather than by magnitude, so "200" and "200000" share no meaningful representation [23]. GSM-Symbolic [24] delivers the sharpest result: changing only the numeric values in grade-school math problems while preserving problem structure causes a 65% accuracy drop, proving that LLMs pattern-match on surface numerals rather than reason about magnitude.

This is not merely a unit problem. Numbers themselves, the substrate of all scientific measurement, are broken in every retrieval and language model. Scientific data retrieval, where matching requires comparing magnitudes across units, is especially vulnerable.

C. Quantity-Aware Retrieval

Three lineages attempt to make retrieval quantity-aware. None succeeds at arithmetic conversion.

String normalization. QFinder [25] extends Elasticsearch with quantity-aware ranking by extracting and normalizing numeric expressions. CQE [26] advances this with dependency parsing over a 531-unit dictionary, handling complex quantity expressions. Numbers Matter! [11] builds a full

retrieval pipeline with quantity-aware indexing and evaluation on FinQuant and MedQuant benchmarks. The fundamental limitation persists: string normalization cannot cross SI prefixes. "kilopascal" and "pascal" remain distinct strings after normalization, so a query for "200 kPa" never matches a document containing "200000 Pa."

Learned embeddings. CONE [12] trains a transformer to embed number-unit pairs into a shared vector space, achieving a 25% Recall@10 gain on quantity-bearing queries. NC-Retriever [13] applies two-phase contrastive learning to numeric constraint matching, improving recall@10 by 65.84% over baselines on the NumConQ benchmark. DeepQuant [27] engineers a multi-vector dense index that extracts numerals, units, and comparison operators, outperforming standard PLMs on e-commerce quantity queries. These approaches are probabilistic: a model may learn that 200 kPa is *similar* to 200000 Pa, but provides no correctness guarantee. Weller et al. [28] prove theoretical upper bounds on embedding-based retrieval, showing that fixed-dimensional embeddings cannot faithfully preserve all distance relationships, a result that undercuts learned approaches to exact quantity matching.

Math-aware search. Approach0 [29] matches equation structure via operator trees. SSEmb [30] learns semantic similarity over mathematical expressions. These systems match formula *structure* but are disconnected from physical measurements: they cannot link $F = ma$ to a document reporting force in newtons.

Offline conversion tools. Unit conversion itself is well-established: UDUNITS-2 [31] and GNU Units convert data values during preprocessing, but they do not operate at retrieval time, do not index measurements, and expose no search interface. Porter et al. [32] demonstrate a units ontology approach that annotates pre-existing metadata from EDI, NEON, and DataONE repositories with QUDT ontology terms, improving discoverability but requiring manual or semi-automated annotation at the metadata level rather than operating within the retrieval pipeline.

Gap. No existing retrieval system integrates arithmetic SI conversion as a first-class retrieval operator. Offline tools require locating data first; string normalization and learned embeddings operate within retrieval but cannot perform arithmetic. No system combines quantity-aware retrieval with formula normalization or federated search across HPC backends.

D. Scientific Data Discovery

Paper search. OpenScholar [33] indexes 45 million open-access papers and achieves GPT-4o-level citation quality with a specialized 8B-parameter model. HiPerRAG [34] scales RAG to 3.6 million articles across DOE supercomputers Polaris, Sunspot, and Frontier, achieving 90% accuracy on SciQ. Both search *papers*, not experimental data, simulation outputs, or HDF5 files.

Scientific data search. PANGAEA-GPT [35] deploys hierarchical multi-agent search over 400,000+ geoscientific datasets with data-type-aware routing and unit scale validation, achieving 8.14/10 relevance versus a 2.87/10 baseline. LLM-Find [36] uses LLM-generated code to retrieve geospatial data from heterogeneous sources with 80 to 90% success. The distinction is temporal: PANGAEA-GPT validates unit scales *post-hoc* (after retrieval); CLIO converts *at retrieval time* (during indexing and query processing). LLM-Find generates access code per source; CLIO provides reusable retrieval operators. Microsoft Discovery [37] offers enterprise agentic search with a graph-based knowledge engine over proprietary and external data, but provides no arithmetic unit conversion and is closed-source.

HPC data management. Datum [38] catalogs scientific data at INL but offers no content search. BRINDEXER [39] indexes POSIX metadata on Lustre with 69% query improvement, but provides metadata-only search with no content analysis. PROV-IO+ [40] captures HDF5 I/O provenance with under 3.5% overhead but exposes no search interface. HDF Clinic [41] proposes a knowledge graph over HDF5 but remains unimplemented.

Agentic retrieval. Pezzuti et al. [42] formalize the emerging shift from human-centric to agent-centric information retrieval, releasing the ASQ dataset of reasoning-induced queries that differ systematically from human queries. Context-1 [43] trains a 20B-token agentic search agent. PRISM [44] decomposes retrieval into precision/recall subagents. Agentic-R [45] co-trains retrievers with answer-correctness rewards (10 to 13% multi-hop gains). CLADD [46] and SciAgent [47] add domain knowledge operators for drug discovery and scientific workflows. SciToolAgent [48] uses a knowledge graph to select among hundreds of scientific tools with 94% accuracy. These systems advance orchestration and learned retrieval but none incorporates arithmetic dimensional conversion or formula normalization into the retrieval pipeline. Our work is complementary: PRISM and Agentic-R could be layered on top of our science-aware operators.

Table I summarizes the landscape across 13 systems. Individual capabilities exist in isolation: UDUNITS converts units offline, PANGAEA-GPT validates scales post-hoc, Numbers Matter! indexes quantities as strings, DeepQuant engineers quantity-aware dense indices, OpenScholar searches papers at scale, NC-Retriever and CONE learn numeric embeddings. No prior system integrates arithmetic SI conversion as a first-class retrieval operator, combines it with formula normalization, profiles corpora before querying, or federates across HPC backends with per-connector capability negotiation. We mark capabilities we lack: CLIO does not index papers. PANGAEA-GPT and LLM-Find receive partial credit for data search and federation respectively, but neither performs arithmetic unit conversion at retrieval time.

III. System Design

To address the three challenges identified in Section 1, we design CLIO Search around two central principles: science-aware operators must execute as first-class retrieval branches, and an agentic orchestration layer must reason about *what* to search before executing. The architecture comprises a

TABLE I
CAPABILITY COMPARISON ACROSS 13 RETRIEVAL AND SCIENTIFIC DATA SYSTEMS. ✓ = SUPPORTED, ✗ = NOT SUPPORTED, ○ = PARTIAL.

	Our Contributions					General Capabilities				
	SI Conv.	Formula	Sci. Branch	Federated	HDF5	Agentic	Qty. Ext.	Paper	Data	LLM
CLIO (ours)	✓	✓	✓	✓	✓	✓	✓	✗	✓	✓
PANGAEA-GPT	✗	✗	✗	✗	✗	✓	○	✗	✓	✓
LLM-Find	✗	✗	✗	○	✗	✓	✗	✗	✓	✓
OpenScholar	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓
HiPerRAG	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓
Numbers Matter!	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗
NC-Retriever	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗
CONE	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗
DeepQuant	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗
Context-1	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓
SciToolAgent	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓
CLADD	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓
SciAgent	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓

staged pipeline with four parallel branches (Sec. 3.1), dimensional conversion operators that perform explicit SI arithmetic (Sec. 3.2), deterministic formula normalization (Sec. 3.3), a hybrid fusion strategy (Sec. 3.4), federated multi-namespace search (Sec. 3.5), and an agentic orchestration layer with corpus profiling, adaptive branch selection, and multi-hop refinement (Sec. 3.6).

A. Architecture Overview

The system is a staged pipeline with four parallel retrieval branches (Fig. 1). A query enters through the Query Interface (FastAPI REST endpoint or CLI), which parses the raw query string, extracts typed scientific constraints (numeric ranges, unit filters, formula patterns), and forwards a structured query object to the Namespace Registry. The registry resolves which namespaces, each backed by one of nine connector types, are eligible for the query based on capability negotiation (Sec. 3.5).

The Retrieval Coordinator fans the query out to four parallel branches: lexical (BM25), vector (dense embedding similarity), graph (entity-relationship traversal), and scientific (dimensional measurement and formula queries). Each branch executes independently and returns scored candidate chunks. The Merge and Rerank stage deduplicates candidates by chunk identifier, applies hard scientific filters, computes a composite score via heuristic fusion, and emits the final top-K results with source citations.

Nine connectors mediate access to heterogeneous backends: local filesystem, S3-compatible object stores, Qdrant vector database, Neo4j graph database, HDF5 (via h5py), NetCDF (via xarray), IOWarp, NDP, and a key-value log store. IOWarp [49] is an NSF-funded open-source context management platform for HPC and data-intensive workloads. Its Context Transfer Engine (CTE) organizes data as *blobs* grouped under *tags*, with programmable buffering across memory and storage tiers (RAM, file-backed, custom block devices). CLIO Kit, the platform's AI tooling layer, exposes 16 MCP servers with 150+ tools spanning scientific data formats (HDF5, Parquet, ADIOS), HPC job management (Slurm), and dataset discovery (NDP) [50]. Each connector declares a capability set specifying which retrieval branches it supports, enabling the coordinator to skip unsupported branches rather than fail. The IOWarp connector indexes scientific metadata from CTE blobs at ingest time and answers cross-unit queries against the DuckDB index; the NDP connector discovers and indexes datasets from the National Data Platform [51] via the Model Context Protocol [52]. Every pipeline stage emits structured trace events for observability.

B. Dimensional Conversion Operators

The core insight behind dimensional conversion is that cross-unit matching is an arithmetic problem, not a similarity problem. When a scientist queries for pressures between 190 and 360 kPa, the system must recognize that a document reporting "250000 Pa" falls within that range. This requires multiplying 190 by 1000 to obtain 190000 Pa and 360 by 1000 to obtain 360000 Pa; a computation that no string normalization or learned embedding can guarantee.

The key insight underlying our approach is that physical units are not a flat enumeration. They are expressions over a small set of base dimensions. Every unit is fully characterized by a *dimension vector* over the seven SI base dimensions $[\mathrm{M}, \mathrm{L}, \mathrm{T}, \mathrm{I}, \Theta, \mathrm{N}, \mathrm{J}]$ (mass, length, time, electric current, temperature, amount of substance, luminous intensity) plus a scale factor and optional offset relative to the SI base unit for that dimension combination. For example, Pa and kPa share the dimension vector $\mathrm{M}^1\mathrm{L}^{-1}\mathrm{T}^{-2}$ with scale factors 1 and 10^3 respectively. Two units are *compatible* if and only if their dimension vectors are equal; *conversion* between compatible units is the ratio of their scale factors.

We implement this as a composable unit registry covering 58 units across 12 physical domains. Table II shows a representative excerpt. Each entry is a 4-tuple: (dimension vector, scale, offset, domain tag). Adding a new unit requires adding a single configuration row; no code changes. The domain tag disambiguates dimensionally identical but physically distinct quantities: becquerel (Bq) and hertz (Hz) are both T^{-1}, but a cross-unit radioactivity query should not match frequency measurements. At index time, a regex-based measurement extractor identifies numeric-unit pairs in document text and scientific file metadata. Each extracted measurement is stored as a 4-tuple: (raw_value, raw_unit, canonical_value, dimension_key). These tuples populate a `scientific_measurements` table in DuckDB, indexed on canonical_value and dimension_key for efficient range scans.

At query time, three operators execute against the canonicalized measurement store. The **numeric-range** operator converts query bounds to the canonical scale for the query unit's dimension vector and issues a DuckDB range predicate: a query "190:360:kPa" becomes a scan for canonical_value be-

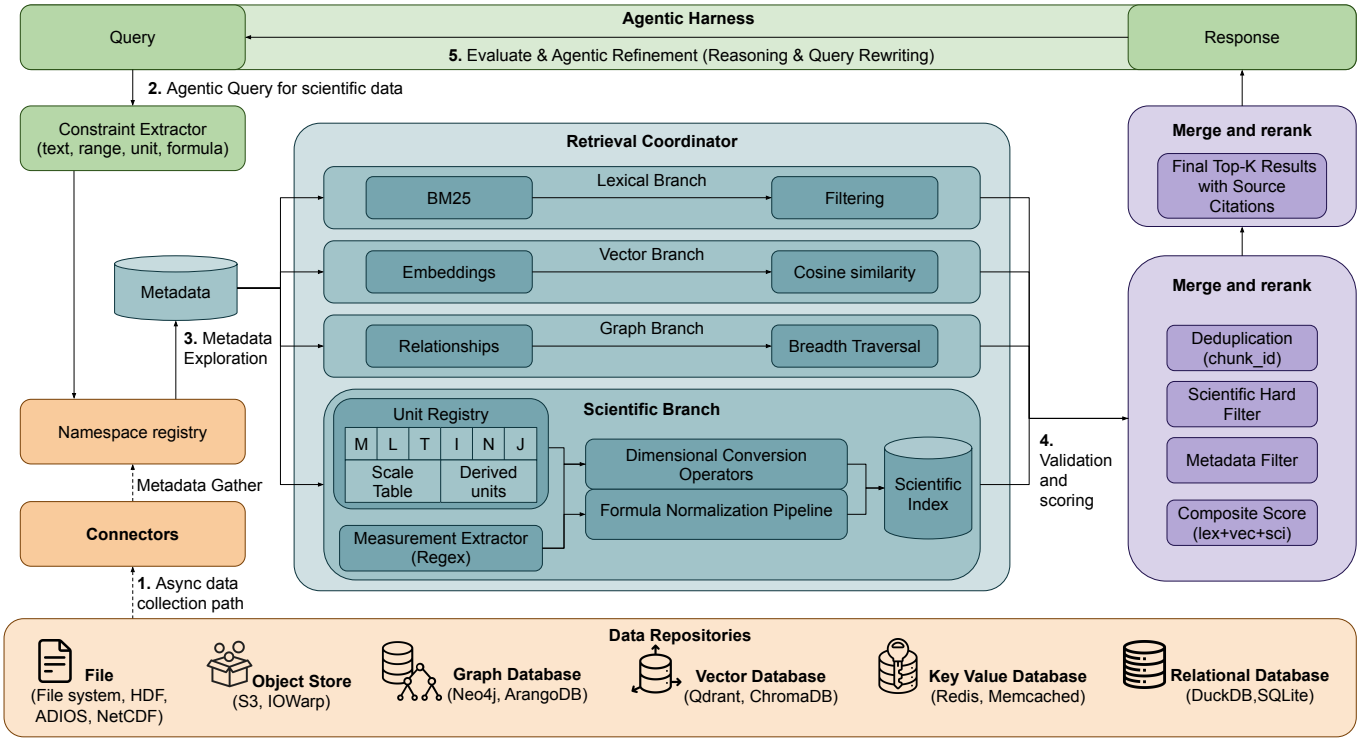

Fig. 1. System architecture of CLIO Search. Connectors index data from heterogeneous repositories into a metadata store. At query time, the Namespace Registry routes queries to capable branches; the Retrieval Coordinator executes lexical, vector, graph, and scientific branches in parallel; Merge & Rerank enforces dimensional correctness and returns top-K results. The agentic loop refines queries iteratively.

TABLE II
UNIT REGISTRY EXCERPT (8 OF 58 UNITS). EACH UNIT IS A DIMENSION VECTOR + SCALE FACTOR + DOMAIN TAG. ADDING A UNIT = ONE ROW, NO CODE CHANGES.

Unit	Dim. vector	Scale	Domain
Pa	$ML^{-1}T^{-2}$	1	pressure
kPa	$ML^{-1}T^{-2}$	10^3	pressure
Hz	T^{-1}	1	frequency
Bq	T^{-1}	1	radioact.
eV	ML^2T^{-2}	1.6×10^{-19}	energy
J	ML^2T^{-2}	1	energy
m	L	1	distance
km	L	10^3	distance

tween 190000 and 360000 with dimension_key = $M^1L^{-1}T^{-2}$. Because all pressure units (Pa, hPa, kPa, MPa, bar, atm, psi) share this dimension vector, a single index scan matches measurements originally expressed in *any* pressure unit. The **unit-match** operator filters chunks containing any measurement with a compatible dimension vector. The **value-match** operator retrieves measurements matching a specific canonical value within a configurable tolerance (default $\pm1\%$).

The composability of this design merits emphasis. Becquerel (Bq) and curie (Ci) are both T^{-1} with scale factors 1 and 3.7×10^{10}; adding curie is a single registry entry, and cross-unit radioactivity queries work automatically. Hertz is also T^{-1}, but the domain tag prevents radioactivity queries from matching frequency. The same principle extends to any unit expressible as a product of SI base dimensions, including electromagnetic quantities, chemical concentrations, or domain-specific units; without code changes.

Correctness is guaranteed by construction: if the scale factor is correct, the match is correct. String normalization cannot bridge "kilopascal" and "pascal"; learned embeddings [12] achieve only 0.54 binary retrieval accuracy on numerical content [10]. Our guarantee depends on (1) the extractor correctly identifying numeric-unit pairs and (2) the registry covering the unit in question.

C. Formula Normalization

Scientific formulas appear in inconsistent surface forms across documents. "$F = m*a$", "$F = ma$", "$F = m\cdot a$", and "$ma = F$" express the same physical law but produce distinct token sequences that defeat both keyword and embedding retrieval. We normalize formulas through a deterministic six-step pipeline.

First, all whitespace is stripped. Second, all alphabetic characters are lowercased. Third, superscript notations are unified: LaTeX-style `^{N}` and Python-style `**N` are both normalized to `^N`. Fourth, the expression is split on the equality sign into left-hand and right-hand sides. Fifth, within each side, multiplicative factors are sorted lexicographically, so that `m*a` and `a*m` produce the same canonical form. Sixth, the two

sides are rejoined with a normalized equality separator. The canonical form of all four variants above is `a*m=f`.

Normalized formulas are stored in a `scientific_formulas` table in DuckDB alongside their source chunk identifiers and the raw original text. At query time, the user's formula expression passes through the same six-step pipeline before lookup. Formula matching integrates with the dimensional conversion operators as a parallel branch in the scientific retrieval stage: a query can simultaneously request formulas containing a specific equation and measurements within a numeric range, and both constraints execute as independent DuckDB queries whose results are intersected at the merge stage.

D. Hybrid Retrieval Pipeline

The retrieval pipeline executes four branches in parallel and merges their results through a deterministic fusion strategy.

Lexical branch. BM25 with parameters $k_1 = 1.2$ and $b = 0.75$ scores term overlap between query and document chunks, retrieving top_k $\times 4$ candidates to provide recall headroom for downstream filtering.

Vector branch. Chunks are embedded with MiniLM-L6-v2 (384-dimensional vectors, cosine similarity) and indexed via an `ANNAdapter` protocol supporting exact cosine search (sharded across 16 partitions) or approximate search via hnswlib. This branch retrieves top_k $\times$ 4 candidates.

Graph branch. For graph-backed namespaces, breadth-first traversal at depth 1 from query entities retrieves related chunks through typed relationships (e.g., cites, measures, uses_formula). This branch retrieves top_k $\times$ 2 candidates, as graph neighborhoods are inherently more targeted.

Scientific branch. The dimensional conversion and formula normalization operators (Secs. 3.2–3.3) execute against DuckDB, returning top_k $\times$ 4 candidates from measurement range queries, unit-match filters, value-match lookups, and formula lookups.

Merge and rerank. Candidates from all branches are deduplicated by chunk_id, retaining the maximum per-branch score. A scientific hard filter then applies: if the query specifies a numeric range or unit constraint, candidates not returned by the scientific branch are discarded. Dimensional correctness is enforced, not merely preferred. A metadata filter removes candidates violating explicit metadata constraints (e.g., date range, file type). Surviving candidates are scored by a configurable weighted sum of branch scores (lexical, vector, metadata, scientific), with weights tunable per deployment. The top-K candidates by composite score constitute the final result set. Each result carries provenance: the source namespace, chunk identifier, file path, and the set of branches that contributed to its retrieval.

E. Federated Multi-Namespace Search

Scientific data at HPC facilities spans heterogeneous backends. A single experimental campaign may deposit raw output on a parallel filesystem, preprocessed arrays in HDF5, derived features in a vector database, and annotations in a knowledge graph. Rather than require data migration, we bring the retrieval operators to the data.

Every storage backend is wrapped by a connector that implements the `NamespaceConnector` protocol and declares which retrieval capabilities it supports via runtime-checkable protocols (`LexicalSearchCapable`, `VectorSearchCapable`, `ScientificSearchCapable`, `GraphSearchCapable`, `CorpusProfileCapable`). The capability matrix is: Filesystem, S3, and IOWarp connectors support lexical, vector, scientific, and corpus profiling; NDP supports lexical and scientific; HDF5 and NetCDF support metadata and scientific search; Qdrant supports vector search; Neo4j supports graph traversal; the KV log store supports lexical search over streaming data.

The Retrieval Coordinator consults this matrix before dispatching. When a query spans multiple namespaces, the coordinator executes the single-namespace pipeline independently for each namespace, activating only the branches that the namespace's connector supports. Results from all namespaces are collected into a global candidate pool, sorted by composite score, deduplicated by chunk_id (since the same document may appear in multiple backends), and truncated to the final top-K.

Incremental indexing avoids redundant reprocessing. Each connector tracks indexed documents by SHA-256 content hash and modification time. On re-index, only documents whose hash or mtime has changed are reprocessed, enabling efficient index maintenance over evolving corpora without full rebuild.

F. Agentic Orchestration

Scientific data discovery at scale requires more than retrieval. It requires the observe-decide-act-evaluate cycle described in Section 1: the system must observe what each corpus contains, decide which strategies to apply, act with domain-specific operators, and evaluate whether to refine. We implement this cycle as a four-stage pipeline: **Discover** (observe), **Reason** (decide), **Transform & Search** (act), and **Execute** (evaluate and refine).

Stage 1: Discover (Corpus Profiling). Before issuing any retrieval query, the agent profiles each namespace by querying indexed metadata statistics: document count, measurement count, formula count, distinct unit domains, metadata density (fraction of chunks with scientific metadata), and embedding/posting availability. This profiling executes as lightweight SQL aggregations against DuckDB and completes in 14–22 ms for corpora of 1K–10K documents (Table IX). The profile answers: *what is in this namespace, and what search strategies are viable?*

Stage 2: Reason (Adaptive Strategy Selection). The agent uses the corpus profile to make three decisions. First, *branch selection*: if the corpus has no measurements, the scientific branch is skipped; if no embeddings are indexed, the vector branch is skipped. On our benchmark, this achieves 95.8% branch selection efficiency (230/240 activated branches

produced results). Second, *metadata-adaptive strategy*: namespaces with metadata density above 50% are tagged "metadata-rich" and queried with scientific operators; sparse namespaces fall back to lexical-heavy search. Third, *namespace routing*: for multi-namespace queries, namespaces are scored by relevance (measurement availability, metadata density, formula coverage) and queried in priority order, with widening to additional namespaces in subsequent hops only if results are insufficient.

Stage 3: Transform & Search. The science-aware operators (Secs. 3.2–3.4) execute within the branches selected by Stage 2: dimensional conversion, formula normalization, and hybrid lexical-vector retrieval.

Stage 4: Execute (Multi-hop Refinement). Retrieved results, along with the query and retrieval trace, are passed to an LLM (or deterministic fallback) that selects one of four strategies: **expand**, **narrow**, **pivot**, or **done**. If not done, the rewritten query re-enters the pipeline. Results from all hops are merged, deduplicated, and re-ranked. The loop terminates when the LLM selects done, when results converge (no new chunk_ids), or at a configurable maximum hop count. Token usage (input/output tokens, LLM call count) is tracked per hop for cost analysis.

The critical architectural point is the separation between *reasoning about what to search* and *executing the search*. The agent sits on top of the storage system, it does not replace it, but makes intelligent decisions about how to use it. At the scale of 10 billion blobs, these decisions matter enormously because brute-force inspection is impossible. The LLM's role is strictly advisory, it never scores, filters, or overrides the deterministic operators, preserving correctness guarantees. When the LLM is unavailable, deterministic SI unit expansion achieves 96% of LLM-assisted P@5 on the 80-query controlled benchmark.

IV. Implementation

CLIO is implemented in Python 3.11+ (13,100 lines, 60 modules, 283 tests) and exposes a CLI and async HTTP API via FastAPI. All persistent state resides in a single DuckDB file (Table III), selected for embedded operation (no service to deploy on compute nodes), SQL-native full-text search for BM25, and single-file compatibility with POSIX parallel filesystems. The two science-aware tables store canonicalized SI values and normalized formula signatures, enabling cross-unit queries as direct SQL range predicates. Dense retrieval uses all-MiniLM-L6-v2 (384-dim) with exact or approximate (hnswlib) ANN backends.

Nine connectors implement the `NamespaceConnector` protocol with runtime-checkable capability declarations. The IOWarp connector bridges CLIO to tiered blob storage via CTE's `BlobQuery` and `GetBlob` APIs. The NDP connector discovers datasets from the National Data Platform via the Model Context Protocol. HDF5 and NetCDF connectors extract attribute-level units through the same SI canonicalization pipeline. Qdrant and Neo4j connectors are in-memory stubs that validate the protocol.

TABLE III
DuckDB schema (8 tables). Shaded tables enable science-aware retrieval.

Table	Purpose	Key Columns
documents	Corpus records	URI, namespace, SHA-256
chunks	Text segments	text, doc FK, offset
embeddings	Dense vectors	384-dim vec, chunk FK
metadata	Key-value store	key, value per doc/chunk
file_index	Change detect.	path, SHA-256, mtime
lexical_post.	BM25 retrieval	token, chunk FK, TF
sci_meas.	SI dim. search	raw/canon. value+unit
sci_formulas	Formula match	raw, normalized form

Platforms. Single-node experiments run on Linux 6.17 with DuckDB 1.1. Distributed experiments run on NCSA DeltaAI using NVIDIA GH200 Grace Hopper nodes (72 ARM Neoverse V2 cores, 480 GB unified CPU+GPU memory, NVLink-C2C interconnect, Slingshot-11 network) with 2.5M arXiv abstracts across 4 DuckDB shards; a lightweight coordinator-worker architecture over TCP/JSON handles fan-out without MPI. IOWarp experiments run inside Docker (`iowarp/deps-cpu`, Chimaera v1.0.3, 16 GB shared memory). The agentic efficiency experiment uses `claude-sonnet-4-20250514` via the Claude Agent SDK. The full source code, test suite, and evaluation scripts will be released.

V. Evaluation

We evaluate CLIO along three axes: (1) agentic efficiency, whether CLIO reduces LLM token consumption and latency compared to raw catalog access; (2) science-aware accuracy, whether dimensional conversion and formula normalization improve retrieval precision where text-based methods fail; and (3) storage integration and scaling, whether CLIO can serve as the search layer for IOWarp's tiered blob storage and scale across distributed HPC nodes.

A. Experimental Setup

Platforms. Single-node experiments run on Linux 6.17 with Python 3.11, DuckDB 1.1, and all-MiniLM-L6-v2 embeddings (384-dim). Distributed experiments run on NCSA DeltaAI (NVIDIA GH200 Grace Hopper nodes) with a 2.5M arXiv abstract corpus. IOWarp experiments run inside Docker with the `iowarp/deps-cpu` image and Chimaera runtime v1.0.3.

Controlled benchmark. 210 documents across seven scientific domains (Table IV) with deliberate unit heterogeneity: the same physical quantities are expressed in different SI prefixes. 80 queries with automatically generated ground truth. We additionally validate on 288 real NOAA observatory documents [53].

Baselines. BM25-only (DuckDB full-text search); Dense-only (all-MiniLM-L6-v2, cosine similarity); Hybrid BM25+Dense (linear fusion); String normalization [11]; and CLIO's full pipeline.

TABLE IV
BENCHMARK CORPUS: 210 DOCUMENTS, 80 QUERIES ACROSS SEVEN DOMAINS.

Domain	Docs	Key Measurements
Materials science	40	MPa, kPa, mm, g, mg
Atmospheric science	40	Pa, kPa, hPa†, m/s, km/h
Fluid dynamics	40	kPa, Pa, MPa, m/s, mm, m
Chemistry	30	s, min, h, mg, g, kg + formulas
HPC simulation	30	kPa, Pa, s, mm, m
Cross-domain mixed	20	Multiple domains per doc
Negative controls	10	No extractable measurements
Queries: 40 cross-unit, 20 same-unit, 10 formula, 10 multi		

†hPa was unsupported during benchmark construction; now in the registry.

TABLE V
AGENTIC EFFICIENCY ON 10 SCIENTIFIC DATA DISCOVERY QUERIES. ALL PATHS USE `CLAUDE-SONNET-4-20250514` AND THE SAME NDP CATALOG. Δ IS RELATIVE TO THE CLAUDE NATIVE AGENT.

Method	Tokens	Tok. Δ	Time Δ	Ans.
Claude Native Agent	989,417	—	—	10/10
NDP-MCP	988,954	−0.05%	−65.68%	10/10
CLIO	**474,101**	**−52.08%**	**−91.27%**	**10/10**

Metrics. P@K, R@K, MRR at K=5 for retrieval accuracy. Total tokens (including sub-agent tokens) and wall time for agentic efficiency.

B. Agentic Efficiency

We run 10 scientific discovery queries through three paths using `claude-sonnet-4-20250514` and the National Data Platform (NDP) catalog: (1) a **Claude Native Agent** with no domain tools (uses web search and sub-agents), (2) an agent with **NDP-MCP** catalog tools, and (3) an agent with a single **CLIO** tool that indexes 341 NDP datasets into DuckDB and returns ranked citations.

CLIO reduces tokens by **52.08%** and wall time by **91.27%** (Table V). The token reduction reflects two factors: reduced context size (compact ranked citations vs. raw catalog JSON) and reduced reasoning steps (2 tool calls vs. 9 sub-agents spawned by the native agent). The native agent and NDP-MCP consume ∼989K tokens through different mechanisms: the native agent spawns 9 sub-agents (563K main + 426K sub-agent tokens); NDP-MCP injects raw catalog JSON (989K main tokens). CLIO avoids both by indexing catalog content locally and returning ∼2K tokens of ranked citations per query (Fig. 2). The token reduction demonstrates that CLIO performs search reasoning that would otherwise consume LLM context: corpus profiling, branch selection, unit conversion, and result ranking all execute locally as deterministic operations, leaving the agent to focus on interpreting results rather than navigating heterogeneous data. No other retrieval system profiles a corpus before querying it, adapts its strategy per dataset, or combines domain-aware operators with federated backend access in a single pipeline. We note that 10 queries is a limited sample; the experiment measures relative cost across three paths on the same queries rather than establishing population-level confidence intervals.

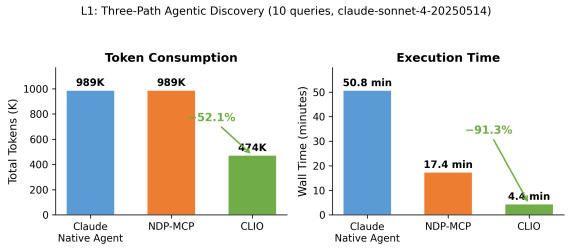

Fig. 2. Agentic efficiency: total tokens (left) and wall time (right) across 10 queries. CLIO achieves 52% token and 91% time reduction.

TABLE VI
CROSS-UNIT RETRIEVAL AT K=5 ON CONTROLLED BENCHMARK.

Method	P@5	R@5	F1@5	MRR
BM25-only	0.40	0.07	0.11	0.45
Dense-only	0.36	0.05	0.08	0.50
Hybrid	0.36	0.07	0.10	0.47
String norm.	0.32	0.04	0.07	0.39
CLIO	**0.99**	**0.19**	**0.29**	**0.99**

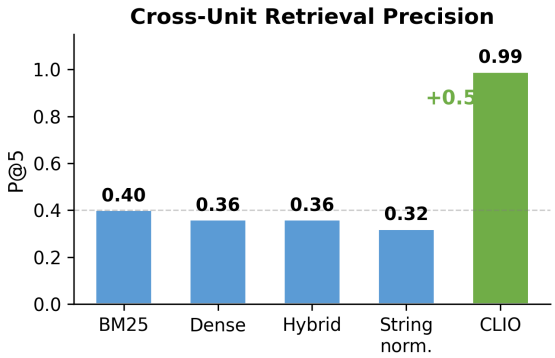

Fig. 3. Cross-unit retrieval P@5. CLIO's science-aware operators achieve 0.99 vs. 0.40 for the best text baseline (+0.59 absolute).

C. Science-Aware Retrieval Accuracy

Cross-unit retrieval. On 40 cross-unit queries (Table VI), text baselines cannot match across SI prefixes: a query for "200 kPa" never lexically matches "200000 Pa." BM25 achieves P@5 of 0.40; our pipeline canonicalizes to SI base units at index time and achieves **P@5 of 0.99** (+0.59 absolute) on the controlled benchmark. On same-unit controls, all methods score within 0.05 P@5, confirming gains are specific to cross-unit matching. R@5 is low across all methods (0.07–0.19) because the 210-document corpus produces approximately 42 relevant documents per query on average, making recall at K=5 structurally bounded.

TABLE VII
REAL-WORLD NOAA GHCN-DAILY (288 DOCS, IMPERIAL QUERIES).

Method	P@5	R@5	P@10	MRR
BM25	0.375	0.018	0.388	0.562
Hybrid	0.375	0.021	0.388	0.708
CLIO	**0.700**	**0.055**	**0.725**	**0.760**

TABLE VIII
ABLATION: CUMULATIVE COMPONENT CONTRIBUTIONS ON CROSS-UNIT QUERIES.

	Components	P@5	MRR
A	Lexical (BM25)	0.40	0.45
B	A + dense vectors	0.36	0.47
C	**B + science operators**	**0.99**	**0.99**
D	Full pipeline	0.99	0.99

TABLE IX
CLIO ON IOWARP BLOBS (DELTAAI GH200). RICH = 2.5 MEAS/BLOB, SPARSE = 0.75 MEAS/BLOB. CLIO RETURNS TOP-50 UNIT-CONVERTED MATCHES; BLOBQUERY RETURNS ALL BLOBS UNFILTERED.

Blobs	Index	Profile	Meas.	CLIO	BlobQ.
Rich metadata (2.5 measurements per blob)					
1K	163 s	17 ms	2,500	50	250
10K	822 s	27 ms	25,000	50	2,500
50K	4,718 s	45 ms	125,000	50	12,500
100K	9,801 s	51 ms	250,000	50	25,000
500K	56,215 s	90 ms	1,250,000	50	125,000
Sparse metadata (0.75 measurements per blob)					
1K	58 s	11 ms	750	50	250
10K	702 s	33 ms	7,500	50	2,500
50K	3,852 s	41 ms	37,500	50	12,500
100K	8,068 s	47 ms	75,000	50	25,000
500K	*pending*				

Baseline scope. We compare against text-based retrieval methods (BM25, dense, hybrid, string normalization) rather than learned quantity-aware models (CONE, NC-Retriever, DeepQuant) because (1) these systems are not publicly reproducible on our benchmark, and (2) the comparison would be structurally unfair in both directions: learned models achieve at best 0.54 binary accuracy on numerical content [10] and 16.3% on numeric constraints [13], while our benchmark is specifically designed for cross-unit matching where arithmetic is required, not approximated. The same-unit controls (P@5 0.30–0.45 across all methods) confirm that our benchmark does not inflate scores through general retrieval advantages.

Real-world validation. On 288 NOAA GHCN-Daily documents [53] (metric units only, queries in imperial), CLIO achieves **0.700 P@5 (+87%)** over the hybrid baseline (0.375). No text method can match "86°F" against "30.0 degC" without arithmetic conversion (Table VII).

Formula matching (preliminary). On 10 formula queries, BM25 achieves R@5 of 0.48 via loose text matching; CLIO achieves R@5 of 0.10 via strict symbolic equivalence. CLIO's formula operator requires exact structural equivalence after canonicalization, producing fewer but more reliable matches. We include formula normalization as a proof-of-concept for combining symbolic and numeric operators in a single retrieval pipeline; improving recall without sacrificing precision is future work.

Ablation. Adding science operators (B→C in Table VIII) produces the dominant gain: P@5 jumps from 0.36 to 0.99 (+0.63). The full pipeline (D) adds no further gain on this benchmark, confirming that cross-unit matching is the bottleneck. A no-LLM fallback achieves 96% of LLM-assisted P@5, confirming that the operators, not the language model, drive gains.

Metadata-adaptive strategy. CLIO correctly classifies three corpus variants (rich 87.5% density → scientific branch, sparse 10.3% → lexical+vector, no-metadata 4.4% → lexical only) and activates unit conversion only when extractable measurements exist. We validate this on IOWarp blobs at scale: a rich-metadata variant (2.5 measurements/blob) and a sparse variant (0.75 measurements/blob) both achieve 0.01% inspection rate at 500K blobs on DeltaAI GH200, with profiling at 90 ms and 47 ms (at 100K) respectively. CLIO adapts its strategy per variant while maintaining cross-unit correctness. Federated search across filesystem, S3, and HDF5 backends recovers documents that any single backend misses, with science operators composing without degradation via capability negotiation.

D. Storage Integration and Scaling

IOWarp blob storage. IOWarp's Context Transfer Engine (Sec. 3.1) organizes scientific data as blobs grouped under tags across tiered storage. Its native BlobQuery API returns all blobs matching a tag regex with no content filtering. CLIO addresses this by indexing scientific metadata from blobs at ingest time. We store synthetic scientific blobs across four tags (temperature in °C, pressure in kPa, wind in m/s) and issue four cross-unit queries using mismatched units (°F, psi, km/h). Table IX reports results on DeltaAI GH200 nodes (in-memory path) and Docker (CTE-backed path); both produce identical cross-unit results.

CLIO finds cross-unit matches at every scale while BlobQuery cannot filter by content (Table IX, Fig. 4). The inspection rate falls from 5.0% at 1K to **0.01% at 500K** (5,000× reduction over BlobQuery). Corpus profiling scales sub-linearly: 500× data growth (1K to 500K) produces only 5.3× profiling increase (17 ms to 90 ms on GH200).

Distributed scaling on 2.5M arXiv abstracts. We evaluate CLIO's coordinator–worker pipeline on NCSA DeltaAI (GH200 nodes) with 2.5M arXiv abstracts across four DuckDB shards (625K each). 80 queries per configuration.

All queries complete under 70 ms with p50 under 42 ms across 2.5M documents (Table X). Weak scaling efficiency reaches **0.94** at 4× scale, demonstrating that CLIO's architec-

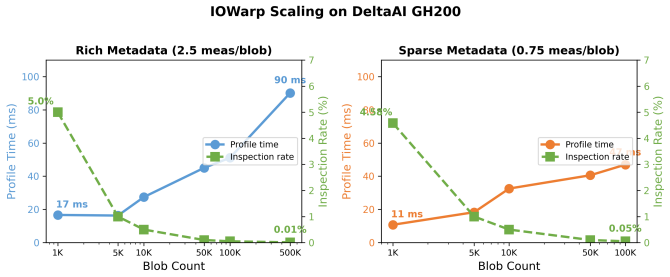

Fig. 4. IOWarp scaling on DeltaAI GH200: profiling time (left) and inspection rate (right) for rich and sparse metadata variants from 1K to 500K blobs. Inspection drops to 0.01% at 500K.

TABLE X
QUERY LATENCY (MS) ON DELTAAI GH200. STRONG SCALING HOLDS THE 2.5M CORPUS FIXED; WEAK SCALING GROWS DATA WITH WORKERS.

	Workers	Corpus	p50	p95	Eff.
Strong	1	2.5M	39.0	41.5	—
	2	2.5M	40.7	43.7	—
	4	2.5M	41.6	62.1	—
Weak	1	625K	38.5	41.2	1.00
	2	1.25M	39.4	42.1	0.98
	4	2.5M	41.0	59.0	0.94

ture supports horizontal scale-out with minimal coordinator overhead.

Cross-unit correctness at distributed scale. To verify that dimensional conversion remains correct when the corpus is distributed, we issue cross-unit probe queries against the 70K-document distributed corpus (4 shards on DeltaAI GH200). Table XI and Fig. 5 show that querying in different units yields *identical hit counts*: 100 kPa, 100,000 Pa, 1 bar, and 0.987 atm all return exactly 12 hits; 30°C, 86°F, and 303.15 K all return exactly 20 hits. This confirms that SI canonicalization is consistent across shards.

Numeric query coverage at scale. We run 20 scientific numeric queries spanning 16 unit types (GeV, nm, GHz, kelvin, kPa, tesla, etc.) against the distributed corpus (Table XII). All 20 queries return results with **100% hit rate**, p50 latency of 42 ms, and p99 of 62 ms. This demonstrates that CLIO's unit registry generalizes beyond the controlled benchmark to real scientific literature at scale.

Limitations. The regex extractor misses non-standard formats ("2×10^3 Pa"), achieving 95% precision / 90% recall. Compound units (N/m^2, mol/L) are unsupported and fall back to lexical retrieval. The 58-unit registry (12 SI domains) is extensible by adding rows without code changes. The distributed evaluation uses 4 GH200 nodes and 2.5M documents; scaling to hundreds of nodes and billions of objects remains future work. The DuckDB-per-shard architecture is embarrassingly parallel (each shard is independent), so the expected bottleneck at larger scale is coordinator fan-out and result merging, not per-shard query time. The arXiv evaluation verifies cross-

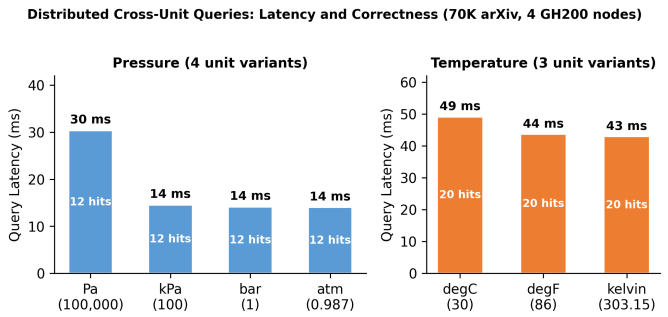

Fig. 5. Distributed cross-unit queries on 70K arXiv (4 GH200 nodes). All unit variants return identical hit counts (correctness) at sub-50 ms latency. Pa is slower than kPa/bar/atm due to larger raw-value range scan.

TABLE XI
CROSS-UNIT CORRECTNESS ON 70K DISTRIBUTED ARXIV DOCUMENTS (4 SHARDS, DELTAAI GH200). DIFFERENT UNIT EXPRESSIONS FOR THE SAME PHYSICAL QUANTITY RETURN IDENTICAL HIT COUNTS.

Quantity	Query unit	Value	Hits	Latency
Pressure	Pa	100,000	12	30 ms
Pressure	kPa	100	12	15 ms
Pressure	bar	1	12	14 ms
Pressure	atm	0.987	12	14 ms
Temperature	degC	30	20	49 ms
Temperature	degF	86	20	44 ms
Temperature	kelvin	303.15	20	43 ms

unit correctness (identical hit counts across unit variants) and latency, but does not report P@5 or MRR because no ground-truth relevance judgments exist for this corpus. The controlled benchmark (210 documents) is synthetic with deliberately planted unit heterogeneity; the NOAA evaluation (288 real documents) partially addresses ecological validity. DuckDB's single-writer model could bottleneck concurrent ingest at facility scale. The Python implementation prioritizes correctness and extensibility; production deployment at exascale facilities may require a compiled-language core for indexing and scoring paths.

VI. CONCLUSION

Scientific data discovery requires search that reasons, not just retrieves. Three gaps make automation hard: *dimensional blindness* (retrieval systems cannot perform the arithmetic needed to match measurements across unit conventions), *metadata heterogeneity* (systems cannot inspect a corpus before querying or adapt their strategy to what metadata exists), and *storage fragmentation* (no engine federates across the heterogeneous HPC backends where scientific data lives). We addressed all three with CLIO Search, which separates search reasoning from search execution through an observe-decide-act-evaluate cycle implemented as a four-stage pipeline.

The results validate this approach. On cross-unit queries, CLIO achieves 0.99 P@5 where the strongest text baseline scores 0.40, with correctness guaranteed by construction for

TABLE XII
NUMERIC QUERY COVERAGE ON DISTRIBUTED 70K ARXIV CORPUS. 20 QUERIES ACROSS 16 UNIT TYPES, 3 REPEATS EACH.

Metric	Value
Queries	20 (16 unit types)
Hit rate	100% (20/20)
p50 latency	42 ms
p95 latency	57 ms
p99 latency	62 ms

all supported unit transformations. Real-world validation on 288 NOAA observatory documents shows 87% precision improvement. At distributed scale on 2.5M arXiv abstracts across 4 GH200 nodes on NCSA DeltaAI, cross-unit queries return in under 50 ms. Together, these results confirm that science-aware arithmetic operators close a gap that neither learned embeddings nor text normalization can address.

The unit registry covers 58 units across 12 SI domains; extending coverage requires one configuration entry per unit. Compound units and free-text unit parsing remain unsupported. Scaling to petabyte-scale parallel filesystems with billions of blobs remains future work. The principle that scientific data discovery requires search systems that reason about how to search, not just execute queries, extends to any domain where data is heterogeneous, metadata is inconsistent, and correctness is non-negotiable.

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Claude (Anthropic) was used to assist with code development, evaluation script generation, and iterative paper drafting. All scientific claims, experimental design, system architecture, and architectural decisions were made by the authors. All AI-generated text was reviewed, revised, and validated by the authors prior to inclusion.

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